



**Pag-CONNECT: A Digital Registration System for
Pag-IBIG Fund Applicants**

College of Computer and Information Sciences
COMP 009 Object-Oriented Programming
BSIT 2-4

Group #7
Yeshua Adrielle S. Arjona
James Adrian M. Escanillas
James Patrick M. Isidro
Raven Joyce L. Rayo
Migel H. Tan

Prof. Severino Bedis Jr.
June 30, 2026

**Table of Contents**

1. Introduction	3
2. Objectives of the System.....	4
3. Scope and Limitations	5
3.1 Scope	5
3.2 Limitations	6
4. System Features	7
5. OOP Concepts Applied.....	10
6. Entity-Relationship Diagram	12
6.1 Database Tables	13
6.2 Primary Keys.....	13
6.3 Foreign Keys	14
6.4 Relationships.....	14
7. System Flowchart and UML Diagram	15
8. Screenshots	17
8.1 Login Screen	17
8.2 Main Menu	18
8.3 Add Record	19
8.4 View Records	21
8.5 Search Record	22
8.6 Update Record	23
8.7 Delete Record	24
8.8 Report Output	25
8.9 Database Tables	26
9. Source Code Explanation	30
9.1 Database Connection	30
9.2 Main Menu	31
9.3 CRUD Operations	32
9.4 OOP Methods	35
9.5 Error Handling	36
10. Conclusion	39



1. Introduction

The Home Development Mutual Fund (HDMF), more commonly known as the Pag-IBIG Fund, is a government-mandated savings program in the Philippines that provides members with access to savings, short-term loans, and housing loan facilities. As part of the membership process, applicants are required to complete the Member's Data Form (MDF), which collects essential personal, demographic, contact, and employment information needed for membership registration. Traditionally, this process involves completing paper-based forms, which can be time-consuming, difficult to manage, and less accessible for individuals who are unable to visit Pag-IBIG offices in person.

To address these challenges, this project introduces Pag-CONNECT: A Digital Registration System for Pag-IBIG Fund Applicants, a database-driven application designed to digitize the Pag-IBIG membership registration process. The system provides a user-friendly Graphical User Interface (GUI) that allows applicants to enter their information electronically while securely storing and managing records through a MySQL database. To improve usability and simplify data entry, the registration process is divided into four modules: Member Information, Current Employment Information, Previous Employment Information, and Heirs/Dependents Information. This modular approach enables users to navigate the registration process more efficiently by organizing related information into manageable sections.

The system is intended for prospective Pag-IBIG members, particularly individuals who have trouble accessing Pag-IBIG branches due to distance, time constraints, or other limitations that make in-person application inconvenient. By transforming the traditional paper-based Member's Data Form into a digital platform, the system minimizes the complexity of manual form completion, improves data organization, and enhances the overall registration experience. Ultimately, Pag-CONNECT serves as a practical solution for streamlining membership registration while demonstrating the application of database management, data modeling, and full-stack system development concepts.



2. Objectives of the System

The general objective of this project is to design and develop a database-driven digital registration system that streamlines the Pag-IBIG Fund membership application process while providing efficient record management capabilities for administrators.

Specifically, the system aims to:

2.1 Digitize the Pag-IBIG Member's Data Form (MDF) by transforming the traditional paper-based registration process into an electronic registration system.

2.2 Provide a user-friendly interface for applicants to conveniently enter and manage their personal, employment, and heir/dependent information through a structured and modular registration process.

2.3 Organize registration data into distinct modules: Member Information, Current Employment Information, Previous Employment Information, and Heirs/Dependents Information, to improve navigation and data entry efficiency.

2.4 Store and manage applicant records using a MySQL database, ensuring organized, accurate, and accessible data management.

2.5 Provide an administrative panel that enables authorized personnel to view, search, update, and manage applicant records efficiently.

2.6 Improve accessibility and convenience for Pag-IBIG applicants, particularly those who may have difficulty visiting physical branches or completing traditional paper-based forms.

2.7 Demonstrate the practical application of database management, relational database design, and full-stack system development concepts through the implementation of a functional registration and management system.



3. Scope and Limitations

This section outlines the coverage and boundaries of the system, specifying its included features, functionalities, and operational constraints.

3.1 Scope

- 3.1.1 Member Registration** - The system allows new applicants to register as Pag-IBIG Fund members by completing four application modules: Member Information, Current Employment, Previous Employment, and Heirs & Dependents. A temporary Pag-IBIG MID number is automatically generated upon starting the registration process and becomes permanent upon successful submission of all required modules.
- 3.1.2 Member Authentication** - Registered members can log in using their Pag-IBIG MID number and password. The system also supports password recovery through security question verification.
- 3.1.3 Member Self-Service Portal** - Logged-in members can view and update their personal information, membership details, employment records, and heirs information through a dedicated dashboard. Members may also edit their complete profile through the Edit Member Information module.
- 3.1.4 Company Management** - The system maintains a master list of companies linked to employment records. Members may select from existing companies or register a new company during the employment module, with automatic company code generation based on the company name initials.
- 3.1.5 Admin Management Portal** - A separate admin login provides authorized personnel access to a management dashboard where they can search, view, edit, and delete records across all four data tables: Members, Current Employment, Heirs, and Companies.
- 3.1.6 Database Integration** - All data is persisted in a MySQL relational database with properly structured tables, foreign key constraints, and cascading operations to maintain data integrity.



3.2 Limitations

- 3.2.1 No Real Pag-IBIG Integration** - The system is a standalone prototype and is not connected to any official Pag-IBIG Fund database or government infrastructure. MID numbers generated are sequential and for simulation purposes only.
- 3.2.2 No Password Encryption** - Passwords are stored in plain text in the database. In a production environment, passwords should be hashed using a secure algorithm such as bcrypt or SHA-256 with salting.
- 3.2.3 Desktop Application Only** - The system is built as a Java Swing desktop application and is not accessible via web browser or mobile device. It requires a local Java runtime environment and an active connection to the MySQL database server.
- 3.2.4 No Contribution or Loan Tracking** - The system covers only the membership registration aspect of Pag-IBIG. Features such as monthly contribution tracking, loan applications, and benefit claims are outside the scope of this project.
- 3.2.5 Local Database Only** - The system connects to a locally hosted MySQL instance. There is no support for cloud-hosted databases, multi-user concurrent access over a network, or data synchronization between multiple installations.



4. System Features

This section presents the key functionalities and capabilities of the Pag-CONNECT Digital Registration System. It highlights the features available to both applicants and administrators, demonstrating how the system supports membership registration, data management, authentication, and record maintenance.

4.1 Member Registration and MID Generation - The system allows new applicants to register as Pag-IBIG Fund members through a guided multi-module registration process. Upon initiating registration, the system automatically queries the database for the last existing MID number and increments it to generate a unique temporary Pag-IBIG MID number displayed throughout the process. The MID becomes permanent only after all required modules are completed and submitted.

4.2 Multi-Module Application Form - The registration process is divided into four modules that the applicant must complete. The Member Information module collects personal details, membership classification, address, contact information, and income data. The Current Employment module captures the applicant's present employer, occupation, employment status, and country of assignment, with companies loaded directly from the database. The Previous Employment module allows applicants to add multiple past employers with corresponding date ranges. The Heirs and Dependents module records the names, relationships, and birthdates of the applicant's designated beneficiaries. Previous Employment, Current Employment, and Heirs are marked as optional modules, while Member Information is required before submission is allowed.

4.3 Report Generation Module - The admin dashboard includes an Application Status Report accessible from the Members tab. The report displays all members in the system alongside a calculated completion percentage based on which modules have been submitted for their MID number. The completion logic checks for the presence of records across the four application tables, `membertable`, `currentemprecordtable`, `prevemtable`, and `heirstable`. Member Information counts as 25%, Current Employment as 25%, Previous Employment as 25%, and Heirs as 25%, yielding a total of 100% for fully registered members. Members who began registration but did not complete all modules are displayed with their partial percentage and flagged as Incomplete, allowing administrators to identify



and clean up abandoned registrations. The report can be filtered and is displayed in a styled table within the admin dashboard.

4.4 Company Master List with Auto Code Generation - The system maintains a master list of companies stored in the database. During the employment modules, applicants may select from existing companies via a dropdown populated from the database. If the applicant's employer is not listed, they may select the "Other" option and provide the company name, address, office assignment, and branch location. The system automatically generates a company code from the initials of the company name and checks for duplicates before saving the new entry.

4.5 Member Login and Authentication - Registered members log in using their Pag-IBIG MID number and password. The system queries the credentials table and grants access only upon a successful match. Invalid attempts display an appropriate error message without disclosing which field was incorrect.

4.6 Password and Security Question Setup - Upon completing all required application modules, the applicant is directed to set up a password for their account. The system validates that the password meets a minimum length requirement and that the confirmation matches. The applicant also selects three distinct security questions and provides answers, which are stored for use in password recovery.

4.7 Password Recovery - Members who forget their password may initiate recovery through the Forgot Password flow. They provide their MID number and answer their three previously set security questions. If all answers match the stored values, the system allows them to reset their password.

4.8 Member Self-Service Dashboard - After logging in, members are presented with a dashboard displaying their MID number and four module cards for viewing their records. Each card opens the corresponding view where the member can read their stored information across all modules.

4.9 Member Information View and Edit - Members can view all their personal, membership, address, contact, and income information in a read-only display. Clicking the Edit button unlocks all fields except the MID number, allowing the member to update their information and save it back to the database.

4.10 Employment and Heirs View - Members can view their current employment record including employer, occupation, employment status, type of work, and country of assignment. Previous employment history is displayed as a list of entries showing the company name and service dates. Heirs and dependents are similarly listed with their relationship and birthdate.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

9

- 4.11 Admin Login Portal** - A separate login page is provided for system administrators, accessible from the main login screen. Admin credentials are stored in a dedicated table independent of the member credentials table, ensuring separation between member and administrative access.
- 4.12 Admin Dashboard with Read, Update, and Delete** - The admin dashboard provides four tabbed views covering Members, Employment, Heirs, and Companies. Each tab displays a searchable and filterable table of all records in the database. Administrators may edit any record through a dialog form or permanently delete records. Deleting a member cascades to remove all associated employment, heirs, and credential records due to foreign key constraints defined in the database schema.



5. OOP Concepts Applied

This section discusses the Object-Oriented Programming (OOP) concepts implemented in the development of the Pag-CONNECT Digital Registration System. It explains how principles such as encapsulation, inheritance, polymorphism, and abstraction were applied to improve code organization, maintainability, and system scalability.

OOP Concept	Explanation
Encapsulation	All model classes such as MemberTable, HeirsTable, CurrentEmpRecordTable, PrevEmpTable, CompanyDetailsTable, and UserCredentials declare their fields as private. Access to these fields is provided exclusively through public getter and setter methods, ensuring that internal data cannot be directly modified from outside the class.
Inheritance	The view and form classes extend JFrame or JPanel from the Java Swing library, inheriting windowing and rendering behavior. For example, MemberInfoFormView extends JPanel, SignInFrame extends JFrame, and AdminDashboard extends JFrame, reusing the layout management, event handling, and painting capabilities of their parent classes without reimplementing them.
Polymorphism	Method overriding is used throughout the UI layer to customize the appearance of Swing components. paintComponent() is overridden in custom JPanel, JButton, JTextField, and JComboBox subclasses to implement the dark glass aesthetic of the application. ActionListener is also implemented as lambda expressions and anonymous classes across multiple components, demonstrating runtime polymorphism through the same interface resolving to different behaviors depending on the component.



Abstraction	The DAO layer abstracts all database operations from the UI layer. Classes such as MemberDAO, HeirsDAO, CurrentEmpDAO, PrevEmpDAO, CompanyDAO, and UserCredentialsDAO expose clean methods like insertMember(), getMemberById(), updateMember(), and deleteMember() without exposing SQL logic to the calling code. The UI only interacts with these high-level methods, remaining independent of database implementation details.
Object Composition	The RegistrationSession class composes a MemberTable object and a UserCredentials object within itself, holding all in-progress registration data in memory during the multi-step signup flow. Rather than passing data between frames through constructors, all forms read from and write to this shared session object, demonstrating object composition as a structural design pattern.
Single Responsibility	Each class is designed with a focused responsibility. Model classes only hold data, DAO classes only handle database operations, frame and panel classes only manage the user interface, and the RegistrationSession class only manages signup state. This separation ensures that changes to one layer do not require changes to others.



6. Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) of the Pag-IBIG Member's Data Form consists of five entities, with an additional associative entity: `membertable`, `heirstable`, `prevemtable`, `currentemprecordtable`, and `companydetailstable`. The `membertable` serves as the central entity, holding the core personal and membership information of each Pag-IBIG member, while the `companydetailstable_has_prevemtable` serves as the associative entity for the many-to-many relationship the two tables have.

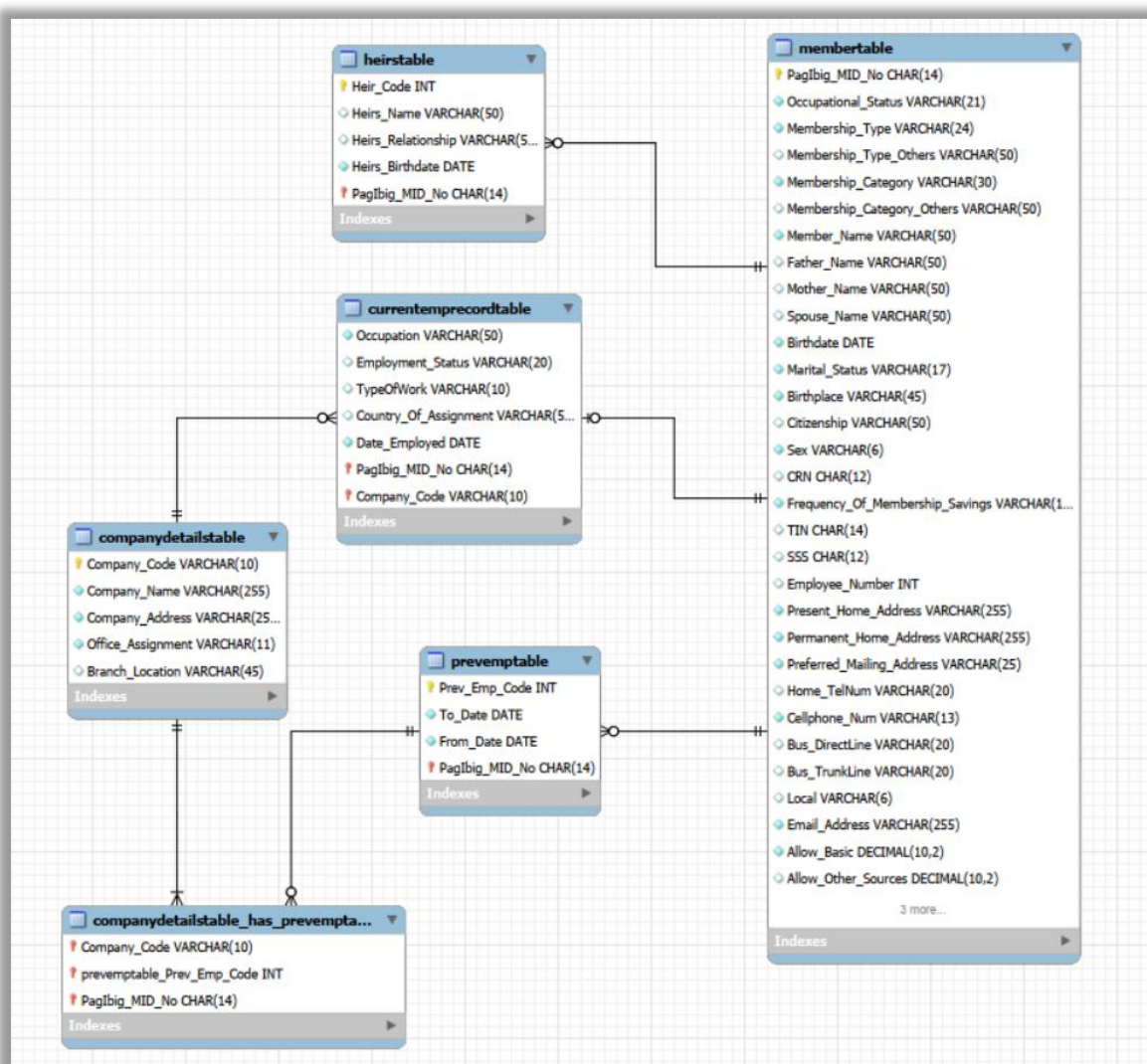


Figure 1. Entity-Relationship Diagram



This section describes the database tables, primary keys, foreign keys, and relationships that form the foundation of the Pag-CONNECT Digital Registration System's data management structure. Note that tables such as usercredentials and admincredentials were not included in the entity-relationship diagram since it's not relevant to the business rules or the application form, but it does exist for the purpose of the system.

6.1 Database Tables

- 6.1.1 membertable** – Serves as the central entity that holds most of the important and required information that a member needs for the application. This is where the Paglbig_MID_No is indicated and used as the primary key for this table, and the foreign key for the remaining ones.
- 6.1.2 currentprecordtable** – Serves as the entity that holds the current employment of the applicant. With respect to the business rules and the real application form, only one current employment is needed in the records, or none.
- 6.1.3 prevemptable** – Serves as the entity holding the previous employments that the applicant has. The applicant can store multiple, or none.
- 6.1.4 heirstable** – Serves as a table for the dependents/heirs of the applicant. Could be multiple or none.
- 6.1.5 companydetailstable** – Serves as the Masterfile of companies existing in the database, together with its name, location, branch, and office assignment.
- 6.1.6 usercredentials** – Serves as the table holding the password of the user with respect to the Paglbig_MID_No.
- 6.1.7 admincredentials** – Serves as the table holding the password and the username of the system's administrator.

6.2 Primary Keys

- 6.2.1 Paglbig_MID_No** – The natural primary key for all tables excluding the companydetailstable. Used as composite for tables that have multiple record functions.
- 6.2.2 Company_Code** – The primary key for the companydetailstable. Populated by the administrator and generated by a method.
- 6.2.3 Heir_Code** – Surrogate primary key for heirstable.
- 6.2.4 Prev_Emp_Code** – Surrogate primary key for prevemptable.



6.3 Foreign Keys

6.3.1 PagIbig_MID_No – Used as foreign key for all tables except the membtable (pk) and companydetailstable.

6.3.2 Company_Code – Used as foreign key for tables such as currenttemprecordtable and prevemptable to include company information.

6.4 Relationships

6.4.1 membtable -> heirstable – Optional-many relationships since a member could have none, or many dependents.

6.4.2 heirstable -> membtable – Mandatory-one relationship since the dependents/heirs should have a designated member.

6.4.3 membtable -> currenttemprecordtable – Optional-one relationship since a member could have one or no current employment in the application.

6.4.4 currenttemprecordtable -> membtable – Mandatory-one relationship since a current employment record must belong to a member.

6.4.5 membtable -> prevemptable – Optional-many relationships since a member could have many previous employment or none.

6.4.6 prevemptable -> membtable – Mandatory-one relationship since a previous employment or multiple, must belong to one member.

6.4.7 companydetailstable -> currenttemprecordtable – Optional-many relationships since a company may reference multiple currently employed records, or none.

6.4.8 currenttemprecordtable -> companydetailstable – Mandatory-one relationship since a current employment needs to reference a company.

6.4.9 companydetailstable -> prevemptable – Optional-many relationships since a company may reference multiple previous employment records, or none.

6.4.10 prevemptable -> companydetailstable – Mandatory-many relationships since multiple previous employments must refer to its respective companies.



7. System Flowchart and UML Diagram

This section presents the flowchart of the Pag-CONNECT Digital Registration System. The flowchart illustrates the sequence of processes, decisions, and interactions performed within the system, providing a visual representation of how users and administrators navigate through various functionalities.

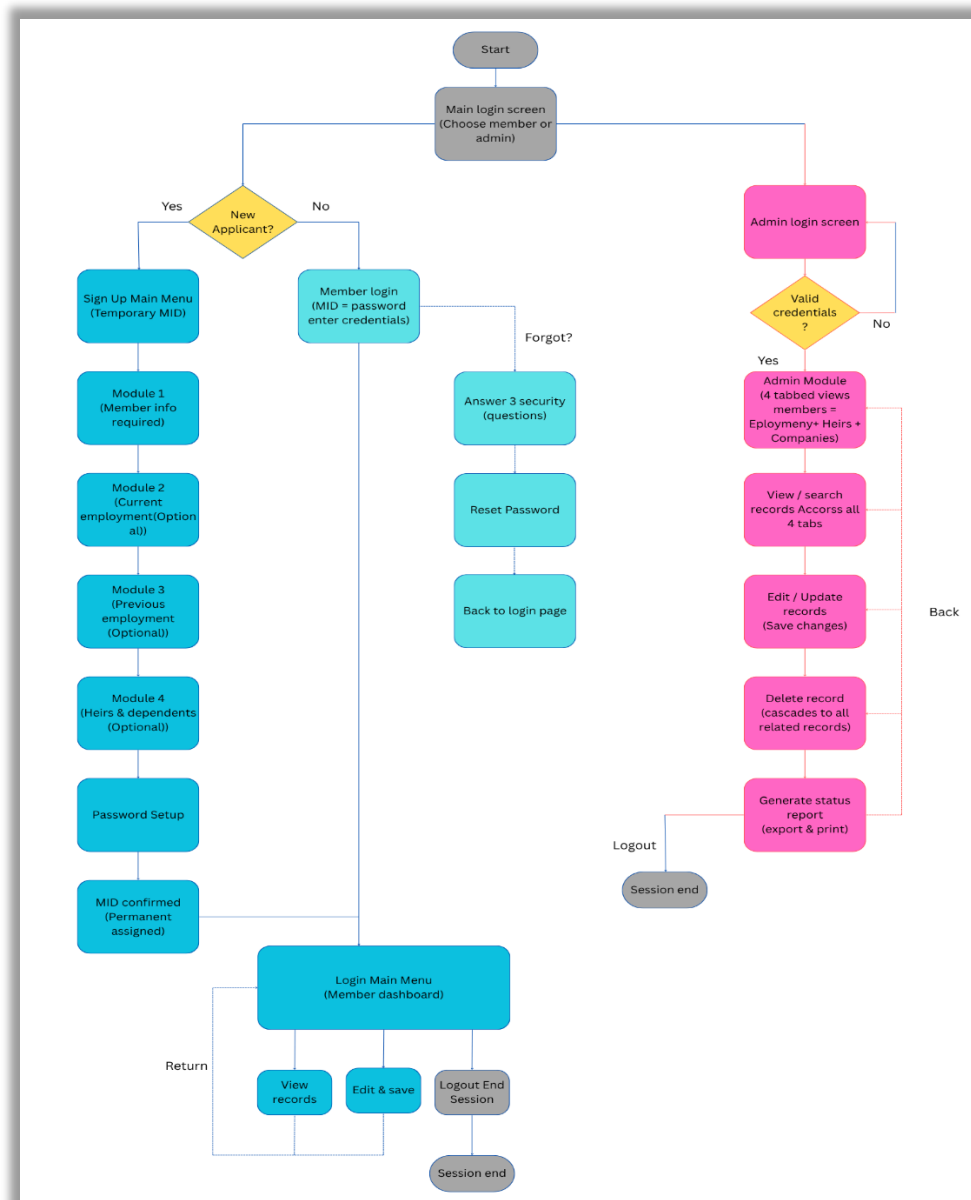


Figure 2. Pag-CONNECT Flowchart



This section presents the UML Diagram of the Pag-CONNECT Digital Registration System. The diagram illustrates the interactions between the system and its users, identifying the major functions available to applicants and administrators and showing how each actor participates in the system's processes.

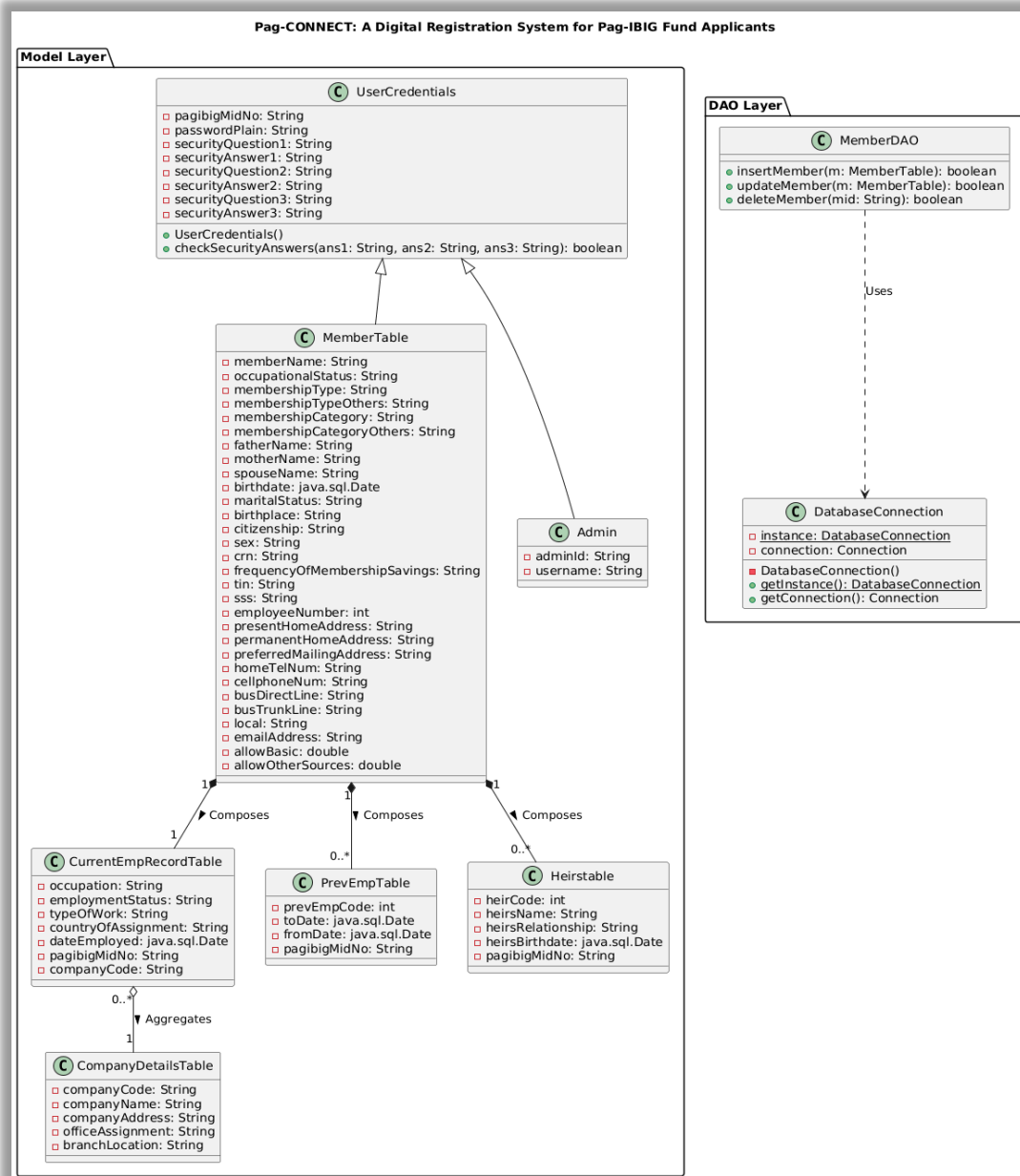


Figure 3. UML Diagram



8. Screenshots

This section showcases the graphical user interface of the Pag-CONNECT Digital Registration System, highlighting the key screens and functionalities available to both members and administrators.

8.1 Login screen

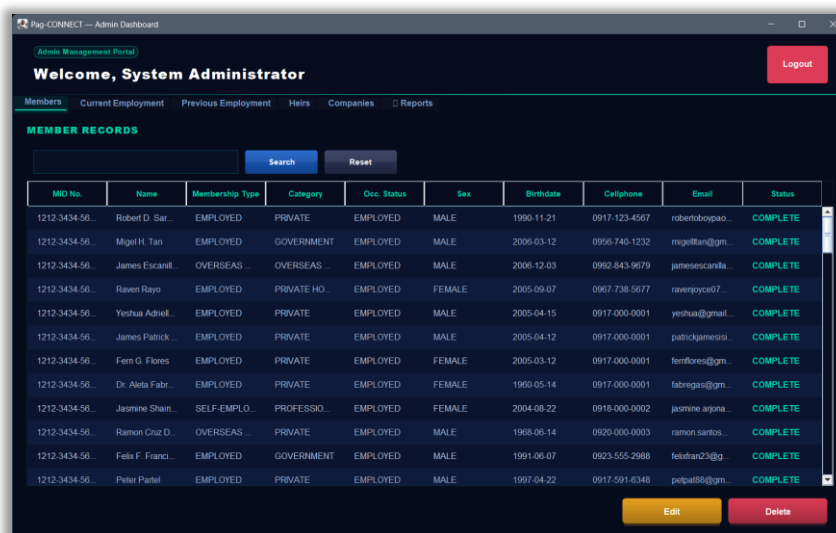
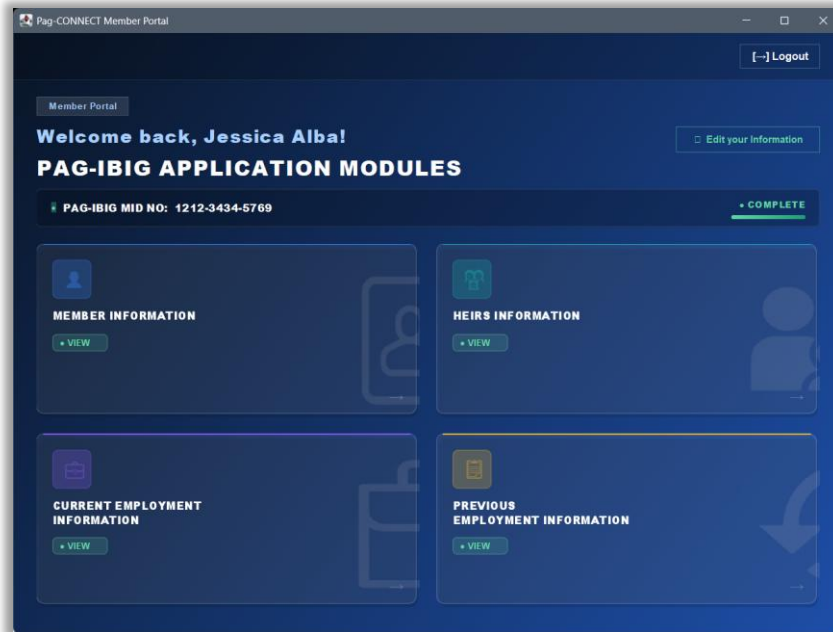
The screenshot shows the 'Pag-CONNECT Member Portal' login interface. The header includes the university logo and the text 'Pag-CONNECT A Digital Registration System for Pag-IBIG Fund Applicants Pag-IBIG Member Portal'. The main content area is split into two sections. On the left, the 'Sign In' section has a form with fields for 'PAG-IBIG 12-DIGIT MID NO.' (containing '1212-3434-5769') and 'PASSWORD' (masked with dots). Below these fields are a 'Remember Me' checkbox and a 'Sign In' button. At the bottom left is a link for 'Admin Access'. On the right, the 'Sign up' section has the text 'Create your account to access the portal.' and a 'Sign Up' button.

The screenshot shows the 'Pag-CONNECT Admin Portal' login interface. The header includes the university logo and the text 'Pag-CONNECT A Digital Registration System for Pag-IBIG Fund Applicants Admin Management Portal'. The main content area features a central 'Admin Sign In' form. Above the form is a 'Restricted Access' warning. The form has fields for 'USERNAME' (containing 'admin') and 'PASSWORD' (masked with dots). Below the fields is a 'Sign In as Admin' button. At the bottom of the form is a link that says '← Back to Member Login'.

Images 1 & 2. Login Screen for User and Admin



8.2 Main menu



Images 3 & 4. Main Menu for User and Admin



8.3 Add record

Pag-CONNECT — Member Information

Member Information Form

Fill out the required information below.

Membership Information

Pag-IBIG MID No. (Temporary): 1212-3434-5770

Membership Type *: Employed

Membership Category *: Government

Occupational Status *: Employed

Frequency of Membership Savings *: Monthly

Personal Information

Member Name *: Member Surname

Father's Name: Father Surname

Mother's Name: Mother Surname

Spouse's Name: Spouse Surname

Birthdate (YYYY-MM-DD) *: 2000-12-12

Birthplace *: Sta. Mesa, Manila

Marital Status *: Single

Sex *: Male

Citizenship *: Filipino

CRN: 40000

Back **Save**

Pag-CONNECT — Current Employment

Current Employment Record

Provide your current employment information.

PAG-IBIG MID NO. (Temporary): 1212-3434-5770

COMPANY *: Accenture - Head Office

COMPANY NAME: Accenture

COMPANY ADDRESS: Taguig City, PH

OFFICE ASSIGNMENT: HEAD OFFICE

BRANCH LOCATION: N/A

OCCUPATION *: Systems Administrator

DATE EMPLOYED (YYYY-MM-DD) *: 2026-01-01

EMPLOYMENT STATUS *: PERMANENT/REGULAR

TYPE OF WORK: Select

COUNTRY OF ASSIGNMENT *: Philippines

Back **Save**



Pag-CONNECT — Heirs Information

Heirs & Dependents

Add your heirs and dependents information.

Heir / Dependent 1 Remove

PAG-IBIG MID NO.	HEIR'S NAME
1212-3434-5770	Member Sibling
RELATIONSHIP	BIRTHDATE (YYYY-MM-DD)
Sibling	2008-12-12

Heir / Dependent 2 Remove

PAG-IBIG MID NO.	HEIR'S NAME
1212-3434-5770	Father Surname
RELATIONSHIP	BIRTHDATE (YYYY-MM-DD)
Parent	1970-12-12

Back **Save**

Pag-CONNECT — Previous Employment

Previous Employment Records

Manage and review your previous employment history.

Previous Employer 1 Remove

PAG-IBIG MID NO. (Temporary)	COMPANY *
1212-3434-5770	NTT Data Philippines - Branch - Bonifacio Global City, Taguig
COMPANY NAME	COMPANY ADDRESS
NTT Data Philippines	Makati City, PH
OFFICE ASSIGNMENT	BRANCH LOCATION
BRANCH	Bonifacio Global City, Taguig
FROM DATE (YYYY-MM-DD) *	TO DATE (YYYY-MM-DD) *
2022-12-11	2025-12-12

Back **Save**

Images 5-8. Create Records



8.4 View records

Pag-CONNECT — Member Record

Edit Member Information

Update your complete member profile and employment history.

Membership Information

Pag-IBIG MID No. 1212-3434-5770	Membership Type Employed
Membership Category Government	Occupational Status Employed
Frequency of Membership Savings Monthly	CRN 12000

Personal Information

Member Name * Member Surname		
Father's Name Father Surname	Mother's Name Mother Surname	Spouse's Name Spouse Surname
Birthdate (YYYY-MM-DD) * 2000-12-12	Birthplace Sta. Mesa, Manila	Marital Status Single
Sex Male	Citizenship Filipino	Employee No. 12012

Government IDs

TIN	SSS No.
-----	---------

[Delete Account](#) [Back](#) [Save All](#)

Pag-CONNECT Member Portal

[--] Logout

Member Portal

Welcome back, Member Surname!

[Edit your Information](#)

PAG-IBIG APPLICATION MODULES

PAG-IBIG MID NO: 1212-3434-5770 [+ COMPLETE](#)

 MEMBER INFORMATION + VIEW	 HEIRS INFORMATION + VIEW
 CURRENT EMPLOYMENT INFORMATION + VIEW	 PREVIOUS EMPLOYMENT INFORMATION + VIEW



The screenshot shows the Pag-CONNECT Admin Dashboard. At the top, there's a header with the university logo and name. Below the header, there's a navigation bar with tabs: Members, Current Employment, Previous Employment, Heirs, Companies, and Reports. The 'Members' tab is selected. The main content area is titled 'MEMBER RECORDS' and features a search bar with 'Search' and 'Reset' buttons. Below the search bar is a table of member records. The table has columns: MID No., Name, Membership Type, Category, Occ. Status, Sex, Birthdate, Cellphone, Email, and Status. The table contains 12 rows of data. At the bottom right of the table, there are 'Edit' and 'Delete' buttons.

MID No.	Name	Membership Type	Category	Occ. Status	Sex	Birthdate	Cellphone	Email	Status
1212-3434-57...	Buenaventura, ...	EMPLOYED	PRIVATE	EMPLOYED	FEMALE	1992-10-31	09177000027	lara.buenavent...	COMPLETE
1212-3434-57...	Hizon, Patrick ...	EMPLOYED	PRIVATE	EMPLOYED	MALE	1987-03-03	09178000028	patrick.hizon@...	COMPLETE
1212-3434-57...	Manalo, Roxan...	EMPLOYED	GOVERNMENT	EMPLOYED	FEMALE	1990-12-25	09179000029	roxanne.manal...	COMPLETE
1212-3434-57...	Tolerino, Ryan...	EMPLOYED	PRIVATE	EMPLOYED	MALE	1985-06-16	09170000030	ryan.tolerino@...	COMPLETE
1212-3434-57...	Ruby Bennett	SELF-EMPLO...	GOVERNMENT	EMPLOYED	FEMALE	2001-09-11	639998299222		INCOMPLETE
1212-3434-57...	Craig Silvenio	EMPLOYED	GOVERNMENT	FIRST TIME J...	MALE	2005-10-02	639770027182		INCOMPLETE
1212-3434-57...	Jana Nicole He...	OVERSEAS ...	PRIVATE	EMPLOYED	MALE	2003-07-21	639928444255	jananicole@gm...	COMPLETE
1212-3434-57...	Mary Grace D...	OVERSEAS ...	PRIVATE HO...	EMPLOYED	FEMALE	2004-02-12	632333333333		INCOMPLETE
1212-3434-57...	Jaya Mikaela ...	OVERSEAS ...	4PS	UNEMPLOYED	FEMALE	2004-01-27	634563636366	jayamikaela@...	COMPLETE
1212-3434-57...	Juan Dela Cruz	EMPLOYED	GOVERNMENT	EMPLOYED	MALE	2002-02-02	639988777777	jdieacruz@gm...	COMPLETE
1212-3434-57...	Jessica Alba	EMPLOYED	PRIVATE	EMPLOYED	FEMALE	2002-12-12	634424242424	jessica@gmail...	COMPLETE
1212-3434-57...	Member Surna...	EMPLOYED	GOVERNMENT	EMPLOYED	MALE	2000-12-12	9999999999	person@gmail...	COMPLETE

Images 9-11. View Records from User Login and Admin Login

8.5 Search record

The screenshot shows the Pag-CONNECT Admin Dashboard with the search function used. The search bar contains the text '1212-3434-5770'. The table below shows the search results, which include the record for 'Member Surna...' with MID No. 1212-3434-5770. The table has columns: MID No., Name, Membership Type, Category, Occ. Status, Sex, Birthdate, Cellphone, Email, and Status. At the bottom right of the table, there are 'Edit' and 'Delete' buttons.

MID No.	Name	Membership Type	Category	Occ. Status	Sex	Birthdate	Cellphone	Email	Status
1212-3434-5770	Member Surna...	EMPLOYED	GOVERNMENT	EMPLOYED	MALE	2000-12-12	9999999999	person@gmail...	COMPLETE

Image 12. Search Record (MID or Name)



8.6 Update record

Pag-CONNECT — Member Information

Member Information

View and manage your member details.

Present Home Address *

Anonas Street, Sta. Mesa, Manila

Permanent Home Address *

Urdaneta City, Pangasinan

Preferred Mailing Address

Present Home Address

Select

Present Home Address

Permanent Home Address

Employer/Business Address

Business Direct Line

012345

Business Trunk Line

01234

Local/Extension

01

Income Information

Basic Allowance *

30000.00

Other Sources

2000.00

Total Monthly Income

32000.00

Back **Save Changes**

Pag-CONNECT — Member Information

Member Information

View and manage your member details.

Membership Information

Pag-IBIG MID No.

1212-3434-5770

Membership Type

EMPLOYED

Membership Category

GOVERNMENT

Occupational Status

EMPLOYED

Frequency of Membership Savings

Quarterly

Success

Member information updated successfully!

OK

Personal Information

Member Name *

Member Surname

Father's Name

Father Surname

Mother's Name

Mother Surname

Spouse's Name

Spouse Surname

Birthdate (YYYY-MM-DD) *

Birthplace

Marital Status

Back **Save Changes**



Pag-CONNECT — Admin Dashboard

Admin Management Portal

Welcome, Sys

Members | Current Employment

MEMBER RECORDS

MID No.	Name	Status
1212-3434-57...	Buenaver...	COMPLETE
1212-3434-57...	Hizon, Pa...	COMPLETE
1212-3434-57...	Manalo, F...	COMPLETE
1212-3434-57...	Tolentino...	INCOMPLETE
1212-3434-57...	Ruby Ben...	INCOMPLETE
1212-3434-57...	Craig Sil...	COMPLETE
1212-3434-57...	Jana Nic...	INCOMPLETE
1212-3434-57...	Mary Gra...	COMPLETE
1212-3434-57...	Jaya Mik...	COMPLETE
1212-3434-57...	Juan Del...	COMPLETE
1212-3434-57...	Jessica A...	COMPLETE
1212-3434-57...	Member S...	COMPLETE

Edit Member — 1212-3434-5770

Member Name * Member Surname

Father's Name Father Surname

Mother's Name Mother Surname

Spouse's Name Spouse Surname

Birthdate (YYYY-MM-DD) 2000-12-12

Birthplace Sta. Mesa, Manila

Marital Status

Sex

Citizenship

CRN 120001

TIN 1200000

SSS 1213100120

Employee Number 12012

ADDRESS & CONTACT

Cancel Save Edit Delete

Info Member updated successfully! OK

Images 13-15. Update Records

8.7 Delete record

Pag-CONNECT — Member Record

Edit Member Information

Update your complete member profile and employment history.

Membership Information

Pag-ID/MID No. 1212-3434-5770

Membership Type Employed

Membership Category Government

Occupational Status Employed

Frequency of Membership Savings Quarterly

CRN 120001

Personal Information

Member Name * Member Surname

Father's Name Father Surname

Mother's Name Mother Surname

Spouse's Name Spouse Surname

Birthdate (YYYY-MM-DD) 2000-12-12

Birthplace Sta. Mesa, Manila

Marital Status Single

Sex Male

Citizenship Filipino

Employee No. 12012

Government IDs

TIN SSS No.

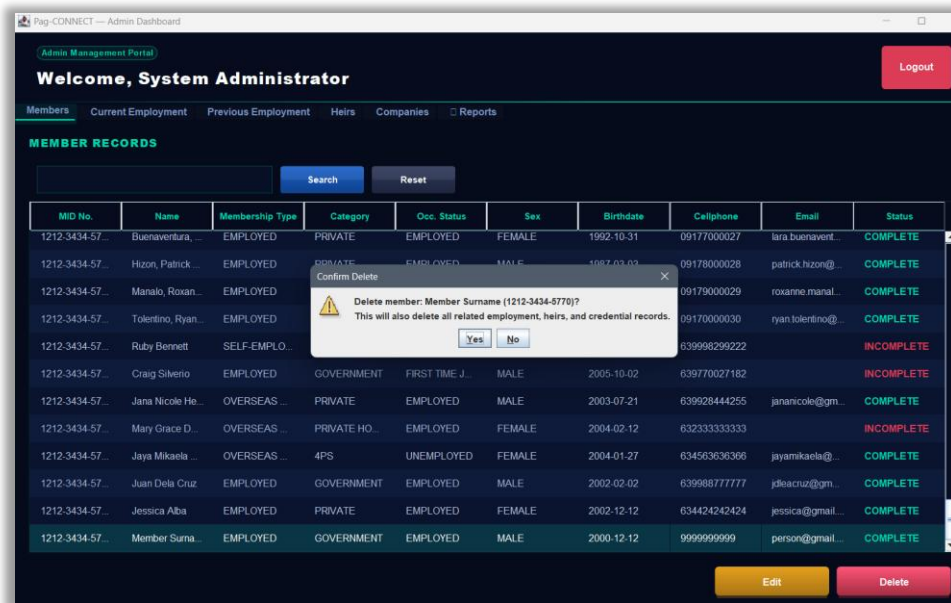
Delete Account Back Save All

Confirm Delete Are you sure you want to delete your account? This action cannot be undone. Yes No



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

25



Images 16 & 17 Delete Record from User Login and Admin Login

8.8 Report output





	A	B	C	D	E	F	G	H	I
1	MID No.	Member N	Member In	Current En	Prev Emp	Heirs	Completion	Status	
2	1212-3434	Robert D. S	YES	YES	NO		75%	IN PROGRESS	
3	1212-3434	Migel H. Ta	YES	YES	YES		100%	COMPLETE	
4	1212-3434	James Esc	YES	YES	NO		75%	IN PROGRESS	
5	1212-3434	Raven Ray	YES	YES	NO		75%	IN PROGRESS	
6	1212-3434	Yeshua Ad	YES	YES	YES		100%	COMPLETE	
7	1212-3434	James Pati	YES	YES	YES		100%	COMPLETE	
8	1212-3434	Fern G. Flo	YES	YES	YES		100%	COMPLETE	
9	1212-3434	Dr. Aleta F	YES	YES	YES		100%	COMPLETE	
10	1212-3434	Catty Hem	YES	YES	YES		100%	COMPLETE	
11	1212-3434	Jasmine St	YES	YES	NO		75%	IN PROGRESS	
12	1212-3434	Ramon Cri	YES	YES	YES		100%	COMPLETE	
13	1212-3434	Felix F. Fra	YES	NO	YES		75%	IN PROGRESS	
14	1212-3434	Peter Part	YES	YES	YES		100%	COMPLETE	
15	1212-3434	Rhian O. Ri	YES	YES	YES		100%	COMPLETE	
16	1212-3434	Diana M. V	YES	YES	YES		100%	COMPLETE	
17	1212-3434	Risa Honth	YES	NO	NO		25%	IN PROGRESS	
18	1212-3434	Amiel And	YES	NO	NO		25%	IN PROGRESS	
19	1212-3434	Alice Guo	YES	YES	NO		75%	IN PROGRESS	
20	1212-3434	Denmar B	YES	NO	NO		50%	IN PROGRESS	
21	1212-3434	Carlos M. I	YES	YES	YES		100%	COMPLETE	
22	1212-3434	Maria Isab	YES	YES	NO		75%	IN PROGRESS	
23	1212-3434	Eduardo B	YES	YES	YES		100%	COMPLETE	
24	1212-3434	Angela C. I	YES	YES	YES		100%	COMPLETE	
25	1212-3434	Juan Paolo	YES	YES	YES		100%	COMPLETE	
26	1212-3434	Maricel P.	YES	NO	NO		50%	IN PROGRESS	
27	1212-3434	Renato O. I	YES	YES	YES		100%	COMPLETE	
28	1212-3434	Sophia Anr	YES	YES	NO		75%	IN PROGRESS	
29	1212-3434	Ernesto V.	YES	NO	YES		75%	IN PROGRESS	

Images 18 & 19 Report Output (GUI & csv file)

8.9 Database tables

presentation

Limit to 1000 rows

```
1. select * from membtable;
```

2

3

PagIbig_MID_No	Occupational_Status	Membership_Type	Membership_Category	Member_Name	Father_Name	Mother_Name	Spouse_Name	Birthdate	Marital_Stal
1212-3434-5753	EMPLOYED	EMPLOYED	GOVERNMENT	Dela Rosa, Carmela Fait...	Timoteo Dela Rosa	Resurreccion Valdez		1994-09-09	SINGLE
1212-3434-5754	EMPLOYED	EMPLOYED	PRIVATE	Santiago, Michael Kevin X.	Clemente Santiago	Adoracion Manalo		1988-01-23	MARRIED
1212-3434-5755	EMPLOYED	EMPLOYED	PRIVATE	Aguilar, Vanessa Nicole Y.	Melanio Aguilar	Corazon Reyes		1995-04-14	SINGLE
1212-3434-5756	EMPLOYED	EMPLOYED	PRIVATE	Dizon, Harold Francis Z.	Proceso Dizon	Visitation Fajardo		1983-07-07	MARRIED
1212-3434-5757	EMPLOYED	EMPLOYED	PRIVATE	Buenaventura, Lara Sop...	Sigfrido Buenavent...	Flordeliza Manalo		1992-10-31	SINGLE
1212-3434-5758	EMPLOYED	EMPLOYED	PRIVATE	Hizon, Patrick Jose BB.	Saturrino Hizon	Nena Tolentino		1987-03-03	SINGLE
1212-3434-5759	EMPLOYED	EMPLOYED	GOVERNMENT	Manalo, Roxanne Claire ...	Epifanio Manalo	Consolacion Padilla		1990-12-25	MARRIED
1212-3434-5760	EMPLOYED	EMPLOYED	PRIVATE	Tolentino, Ryan Joseph ...	Abundio Tolentino	Zenaida Cruz		1985-06-16	SINGLE
1212-3434-5761	EMPLOYED	SELF-EMPLOYED	GOVERNMENT	Ruby Bennett	Father Bennett	Leslie Bennett	Jules Vaughn	2001-09-11	SINGLE
1212-3434-5762	FIRST TIME JOBSSEE...	EMPLOYED	GOVERNMENT	Craig Silverio	wanawina			2005-10-02	SINGLE
1212-3434-5763	EMPLOYED	OVERSEAS FILIPINO WORKER	PRIVATE	Jana Nicole Herrera	Jason Flores	Rona Myla Herrera	Angelo Manalang	2003-07-21	MARRIED
1212-3434-5764	EMPLOYED	OVERSEAS FILIPINO WORKER	PRIVATE HOUSEHOLD	Mary Grace Duterte	Roberto Rodrigo	Sara Rodrigo	Baste Duterte	2004-02-12	MARRIED
1212-3434-5765	UNEMPLOYED	OVERSEAS FILIPINO WORKER	APS	Jaya Mikaela Pineda	Bong Pineda	Pacenda Pineda	Tin	2004-01-27	LEGALLY SEF
1212-3434-5766	EMPLOYED	EMPLOYED	GOVERNMENT	Juan Dela Cruz	Father Dela Cruz	Mother Dela Cruz	Spouse Dela Cruz	2002-02-02	SINGLE
1212-3434-5769	EMPLOYED	EMPLOYED	PRIVATE	Jessica Alba	Father Alba	Mother Alonzo	Spouse Spouse	2002-12-12	MARRIED
1212-3434-5770	EMPLOYED	EMPLOYED	GOVERNMENT	Member Surname	Father Surname	Mother Surname	Spouse Surname	2000-12-12	SINGLE

Image 20. membtable



10. **SELECT * FROM currentprecordtable;**

11

12

PagIbig_MID_No	Company_Code	Occupation	Employment_Status	TypeOfWork	Country_Of_Assignment	Date_Employed
1212-3434-5656	STECH	Network Engineer	CONTRACTUAL	HALL	Philippines	2028-11-11
1212-3434-5657	DICT	Data Scientist	PERMANENT/REGULAR	HALL	Philippines	2024-12-03
1212-3434-5658	CVG	Network Engineer	CASUAL	HALL	Philippines	2028-11-13
1212-3434-5659	JPM	Data Analyst	PROJECT BASED	LAND-BASED	U.S.A	2028-11-14
1212-3434-5660	CIS	Software Developer	PERMANENT/REGULAR	HALL	Philippines	2025-01-01
1212-3434-5669	CIS	Network Engineer	PERMANENT/REGULAR	HALL	Philippines	2024-12-12
1212-3434-5672	ORC	Systems Analyst	PERMANENT/REGULAR	HALL	Philippines	2025-02-01
1212-3434-5673	PUP	Professor III	PERMANENT/REGULAR	HALL	Philippines	2025-02-01
1212-3434-5675	SCPH	Civil Engineer	CONTRACTUAL	HALL	Philippines	2024-03-15
1212-3434-5676	SKEC	Mechanical Engineer	CONTRACTUAL	LAND-BASED	Saudi Arabia	2023-05-01
1212-3434-5677	CSC	Assistant Commissi...	PERMANENT/REGULAR	HALL	Philippines	2025-02-03
1212-3434-5678	HWI	Network Associate	PERMANENT/REGULAR	HALL	Philippines	2020-02-15
1212-3434-5689	HWI	UNEMPLOYED	CASUAL	HALL	Philippines	2025-03-01
1212-3434-5694	PUP	Professor I	PERMANENT/REGULAR	HALL	Philippines	2024-06-01
1212-3434-5697	FOU	Voice	CASUAL	HALL	Philippines	2024-03-12
1212-3434-5698	AWS	Networking	PERMANENT/REGULAR	HALL	Philippines	2025-12-09
1212-3434-5701	CSC	Senior Administrati...	PERMANENT/REGULAR	HALL	Philippines	2018-06-01
1212-3434-5702	ACN	Business Analyst	CONTRACTUAL	HALL	Philippines	2023-03-15

Image 21. currentprecordtable

11. **select * from prevemptable;**

12

13

PagIbig_MID_No	Prev_Emp_Code	Company_Code	To_Date	From_Date
1212-3434-5731	57	BDO	2020-12-07	2018-12-12
1212-3434-5761	58	ORC1	2023-04-12	2022-12-03
1212-3434-5762	59	BDO	2026-04-10	2024-10-03
1212-3434-5764	64	AECO	2024-05-11	2022-12-22
1212-3434-5765	65	AECO	2025-06-23	2024-12-22
1212-3434-5675	68	AECO	2024-02-28	2021-07-01
1212-3434-5675	69	AECO	2024-02-28	2021-07-01
1212-3434-5673	75	ACN	2024-10-27	2022-08-03
1212-3434-5766	88	STECH	2025-09-12	2025-03-19
1212-3434-5766	89	PUP2	2025-03-14	2024-12-03
1212-3434-5766	90	ACN	2022-04-12	2021-12-05
1212-3434-5657	99	PUP	2023-12-11	2022-04-23
1212-3434-5657	100	ACN	2022-04-19	2019-11-24
1212-3434-5657	101	BDO	2011-12-12	2010-12-15
1212-3434-5769	109	ACN	2025-01-02	2024-05-06
1212-3434-5769	110	SMPH	2023-12-12	2022-04-12
1212-3434-5770	121	NTT	2025-12-12	2022-12-11

Image 22. prevemptable



7· `select * from heirstable;`

8

9

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

	PagIbig_MID_No	Heir_Code	Heirs_Name	Heirs_Relationship	Heirs_Birthdate
	1212-3434-5730	56	Alvin P. Dela Rosa	Spouse	1983-08-11
	1212-3434-5730	66	Alvin P. Dela Rosa	Spouse	1983-08-11
	1212-3434-5761	87	Jules Vaughn	Spouse	2001-04-12
	1212-3434-5761	88	Gia Bennett	Sibling	2007-12-06
	1212-3434-5762	89	James Halimaw	Child	2005-12-03
	1212-3434-5763	90	Migel H. Tan	Other	2006-03-12
	1212-3434-5764	101	Hello Duterte	Child	2020-02-12
	1212-3434-5765	102	Mama ko si Pace...	mama ko nga	1992-02-12
	1212-3434-5766	110	Mother Dela Cruz	Parent	1989-12-31
	1212-3434-5766	111	Father Dela Cruz	Parent	1987-05-12
	1212-3434-5766	112	Spouse Dela Cruz	Spouse	2006-12-12
	1212-3434-5769	283	Jonathan Henry...	Sibling	2005-12-04
	1212-3434-5769	284	Josiah Alba	Legal Heir	2002-08-09
	1212-3434-5769	285	Jinrho Cruz	Sibling	2007-12-12
	1212-3434-5769	286	Jerico Alba	Legal Heir	2007-12-12
	1212-3434-5770	307	Member Sibling	Sibling	2008-12-12
	1212-3434-5770	308	Father Surname	Parent	1970-12-12
	NULL	NULL	NULL	NULL	NULL

Image 23. heirstable

14· `select * from companydetailstable;`

15

16

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Company_Code	Company_Name	Company_Address	Office_Assignment	Branch_Location
	ACN	Accenture	Taguig City, PH	HEAD OFFICE	NULL
	AECO	AECOM Philippines	Pasig City, PH	BRANCH	Ortigas Center, Pasig City
	APP	Apple Inc	California, USA	HEAD OFFICE	NULL
	AWS	Amazon Web Services	Quezon City, PH	HEAD OFFICE	NULL
	AWS1	Amazon Web Services	Quezon City, PH	BRANCH	Ortigas, Pasay City
	BDO	Banco de Oro Unibank	Makati City, PH	HEAD OFFICE	NULL
	BPI	Bank of the Philippine Islands	Quezon City	HEAD OFFICE	NULL
	BPI1	Bank of the Philippine Islands	Quezon City	BRANCH	Valenzuela City
	CIS	Cisco Philippines	Taguig City, PH	HEAD OFFICE	NULL
	CSC	Civil Service Commission	Quezon City, PH	HEAD OFFICE	NULL
	CVG	Converge	Pasig City, PH	HEAD OFFICE	NULL
	DICT	Department of Information ...	Quezon City, PH	HEAD OFFICE	NULL
	DOH	Department of Health	Manila City, PH	HEAD OFFICE	NULL
	DOST	Department of Science and ...	Taguig City, PH	HEAD OFFICE	NULL
	FOU	Foundever	Ortigas, Pasig City	HEAD OFFICE	NULL
	GCH	GCash (Mynt)	Taguig City, PH	HEAD OFFICE	NULL
	GLO	Globe Telecom	Taguig City, PH	HEAD OFFICE	NULL
	HWI	Huawei Philippines	Taguig City, PH	BRANCH	Quezon City

Image 24. companydetailstable



16• `select PagIbig_MID_No, Password`
17 `from usercredentials;`
18

Result Grid

PagIbig_MID_No	Password
1212-3434-5752	adminuser
1212-3434-5753	adminuser
1212-3434-5754	adminuser
1212-3434-5755	adminuser
1212-3434-5756	adminuser
1212-3434-5757	adminuser
1212-3434-5758	adminuser
1212-3434-5759	adminuser
1212-3434-5760	adminuser
1212-3434-5761	adminuser
1212-3434-5762	adminuser
1212-3434-5763	adminuser
1212-3434-5764	adminuser
1212-3434-5765	adminuser
1212-3434-5766	adminuser
1212-3434-5769	adminuser
1212-3434-5770	adminuser

Image 25. usercredentials Table

16• `select * from admincredentials;`
17
18

Result Grid

Admin_ID	Username	Password	Admin_Name
1	admin	admin123	System Administrator

Image 26. admincredentials Table



9. Source Code Explanation

9.1 Database Connection

The system establishes its connection to the MySQL database through a centralized DatabaseConnection.java class located in the database package. This class contains a single static method getConnection() that calls DriverManager.getConnection() with the configured URL, username, and password. By centralizing the connection login in one place, every DAO class in the system can obtain a fresh connection by calling DatabaseConnection.getConnection() without duplicating the connection string or credentials anywhere else. Each DAO method opens a connection at the start of its operation and closes it automatically using a try-with-resources block, ensuring that connections are never left open even if an exception occurs during the database operation.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DatabaseConnection {

    private static final String URL      = "jdbc:mysql://localhost:3306/pagibig";
    private static final String USERNAME = "root";
    private static final String PASSWORD = "adminuser";

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USERNAME, PASSWORD);
    }

    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        try (Connection conn = getConnection()) {
            System.out.println(x: "Connected to MySQL Successfully!");
        } catch (SQLException e) {
            System.out.println(x: "Connection Failed!");
            e.printStackTrace();
        }
    }
}
```

Image 27. Database Connection Source Code

The database is named pagibig and is hosted on a local MySQL server. The connection is tested by running the main() method of DatabaseConnection, which prints a success or failure message to the console.



9.2 Main Menu

The main menu for members is implemented in `SignInFrame.java`, which extends `JFrame` and serves as the central navigation hub after a successful login. The frame receives the authenticated MID number as a constructor argument and immediately queries the database through `MemberDAO.getMemberById()` to retrieve the member's name for the personalized welcome label.

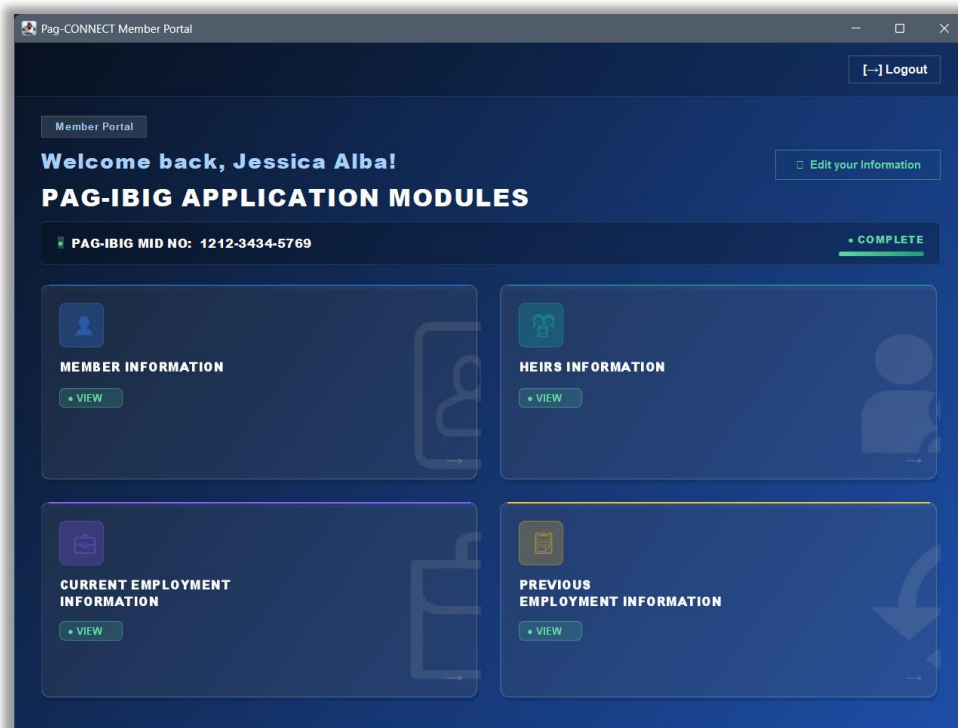


Image 28. Main Menu (Login)

The menu is built as a two-by-two grid of `DarkModuleCard` buttons, each representing one of the four application modules. Each card is a custom inner class that extends `JButton` and overrides `paintComponent()` to render the glass-style card appearance. Clicking a card disposes the current frame and opens the corresponding view class, `MemberInfoFormView`, `CurrentEmpFormView`, `PrevEmpFormView`, or `HeirsFormView`, passing the logged-in MID to maintain session context across all navigation actions.



9.3 CRUD Operations

All create, read, update, and delete operations are handled by the DAO classes in the dao package. Each DAO class is responsible for exactly one database table and exposes clearly named methods for each operation.

```
// --- CREATE ---
public boolean insertMember(MemberTable member) {
    String sql = "INSERT INTO membtable ("
        + "PagIbig_MID_No, Occupational_Status, Membership_Type, "
        + "Membership_Category, Member_Name, Father_Name, "
        + "Mother_Name, Spouse_Name, Birthdate, Marital_Status, Birthplace, Citizenship, "
        + "Sex, CRN, Frequency_Of_Membership_Savings, TIN, SSS, Employee_Number, "
        + "Present_Home_Address, Permanent_Home_Address, Preferred_Mailing_Address, "
        + "Home_TelNum, Cellphone_Num, Bus_DirectLine, Bus_TrunkLine, Local, "
        + "Email_Address, Allow_Basic, Allow_Other_Sources, Total_Mo_Income, Application_Status"
        + ") VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";
```

Image 29. Create Operation

Create operations use PreparedStatement with positional parameters to safely insert records. For example, MemberDAO.insertMember() builds an INSERT statement with thirty parameters mapped to the fields of a MemberTable object. HeirsDAO.insertHeir() similarly inserts a single heir record. All insert methods return a boolean indicating whether the operation affected at least one row.

```
// --- READ ---
public MemberTable getMemberById(String pagIbigMIDNo) {
    String sql = "SELECT * FROM membtable WHERE PagIbig_MID_No = ?";
    try (Connection conn = DatabaseConnection.getConnection();
        PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setString(parameterIndex: 1, pagIbigMIDNo);
        ResultSet rs = ps.executeQuery();
        if (rs.next()) return mapRow(rs);
    } catch (SQLException e) {
        System.err.println("[MemberDAO] getMemberById error: " + e.getMessage());
    }
    return null;
}
```

Image 30. Read Operation

Read operations use either a Statement for fetching all records or a PreparedStatement with a WHERE clause for fetching by identifier. The mapRow() private helper method in each DAO reads column values from the ResultSet and populates the corresponding model object, keeping the mapping logic in one place.



```
// — UPDATE —
public boolean updateMember(MemberTable member) {
    // Fixed: Added Application_Status column modifier to the set parameters
    String sql = "UPDATE membtable SET "
        + "Occupational_Status = ?, Membership_Type = ?, "
        + "Membership_Category = ?, Member_Name = ?, "
        + "Father_Name = ?, Mother_Name = ?, Spouse_Name = ?, Birthdate = ?, "
        + "Marital_Status = ?, Birthplace = ?, Citizenship = ?, Sex = ?, CRN = ?, "
        + "Frequency_Of_Membership_Savings = ?, TIN = ?, SSS = ?, Employee_Number = ?, "
        + "Present_Home_Address = ?, Permanent_Home_Address = ?, "
        + "Preferred_Mailing_Address = ?, Home_TelNum = ?, Cellphone_Num = ?, "
        + "Bus_DirectLine = ?, Bus_TrunkLine = ?, Local = ?, Email_Address = ?, "
        + "Allow_Basic = ?, Allow_Other_Sources = ?, Total_Mo_Income = ?, "
        + "Application_Status = ? "
        + "WHERE PagIbig_MID_No = ?";
```

Image 31. Update Operation

Update operations build an UPDATE statement with SET clauses for every editable field and a WHERE clause on the primary key. MemberDAO.updateMember() for example updates all thirty editable columns of the member record identified by the MID number. The admin dashboard additionally uses CurrentEmpDAO.updateCurrentEmp(), HeirsDAO.updateHeir(), and CompanyDAO.updateCompany() for their respective tables.

```
// — DELETE —
public boolean deleteMember(String pagIbigMIDNo) {
    String sql = "DELETE FROM membtable WHERE PagIbig_MID_No = ?";
    try (Connection conn = DatabaseConnection.getConnection();
        PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setString(parameterIndex: 1, pagIbigMIDNo);
        return ps.executeUpdate() > 0;
    } catch (SQLException e) {
        System.err.println("[MemberDAO] deleteMember error: " + e.getMessage());
        return false;
    }
}
```

Image 32. Delete Operation

Delete operations use a simple DELETE statement with a WHERE clause on the primary key. Because the database schema defines cascading deletes on all foreign keys referencing membtable, deleting a member through MemberDAO.deleteMember() automatically removes all associated current employment, previous employment, heirs, and credentials records without requiring explicit deletion calls for each table.



```
// --- GENERATE NEXT MID ---  
public String generateNextMID() {  
    String sql = "SELECT PagIbig_MID_No FROM membtable ORDER BY PagIbig_MID_No DESC LIMIT 1";  
    try (Connection conn = DatabaseConnection.getConnection();  
        Statement stmt = conn.createStatement();  
        ResultSet rs = stmt.executeQuery(sql)) {  
        if (rs.next()) {  
            String lastMID = rs.getString(columnLabel: "PagIbig_MID_No");  
            String[] parts = lastMID.split(regex: "-");  
            int lastSegment = Integer.parseInt(parts[2]);  
            lastSegment++;  
            return parts[0] + "-" + parts[1] + "-" + String.format(format: "%04d", lastSegment);  
        }  
    } catch (SQLException e) {  
        System.err.println("[MemberDAO] generateNextMID error: " + e.getMessage());  
    }  
    return null;  
}
```

Image 33. Auto-Generated MID Number

MID Generation is implemented as a special read operation in MemberDAO.generateNextMID(). It queries the highest existing MID number, splits it by the dash separator, increments the last four-digit segment, and returns the formatted result. This ensures each new registration receives a sequentially unique identifier.



9.4 OOP Methods

The project is organized into five distinct packages, each serving a clear architectural role.



Image 34. models Package

The models package contains plain Java classes that represent database table rows. MemberTable, HeirsTable, CurrentEmpRecordTable, PrevEmpTable, CompanyDetailsTable, and UserCredentials each declare private fields with public getters and setters. These classes are purely data carriers with no business logic or database awareness.

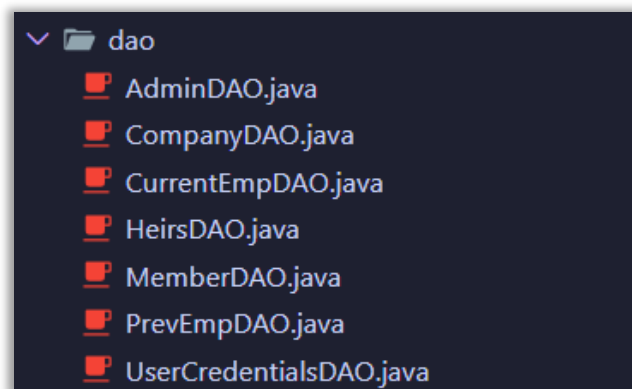


Image 35. dao Package

The dao package contains the data access layer. Each class corresponds to one database table and encapsulates all SQL for that table. The DAO classes depend on the model classes and the DatabaseConnection class but have no knowledge of the UI layer.

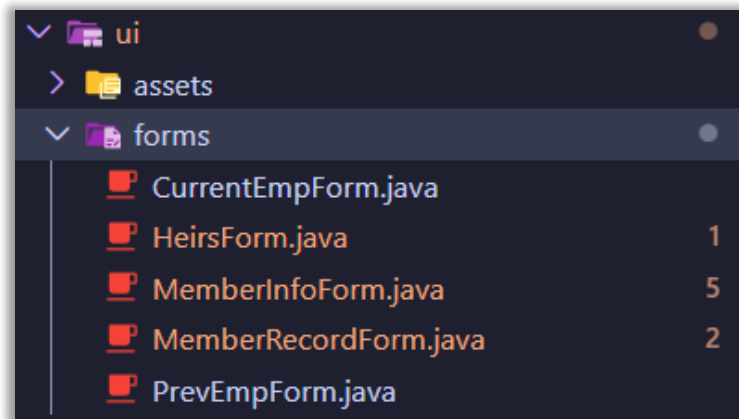


Image 36. forms Package

The ui.forms package contains form panels used during the registration signup flow. MemberInfoForm, CurrentEmpForm, PrevEmpForm, and HeirsForm each extend JPanel and are embedded inside JFrame windows opened from SignUpFrame. MemberRecordForm is the combined edit form accessible from the member dashboard.

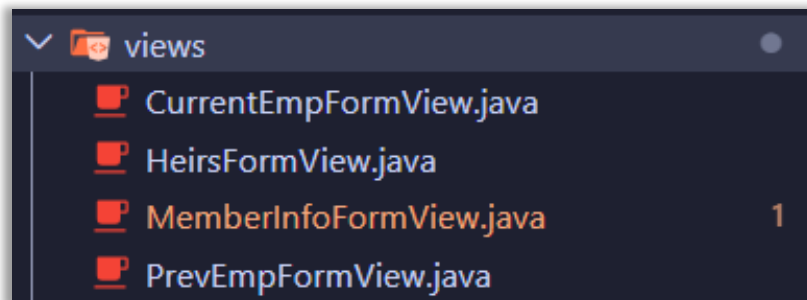


Image 37. views Package

The ui.views package contains read-only and editable view panels displayed after login. MemberInfoFormView, CurrentEmpFormView, PrevEmpFormView, and HeirsFormView each extend JPanel, load data from the database on construction, and display it in a locked state that can be unlocked for editing.

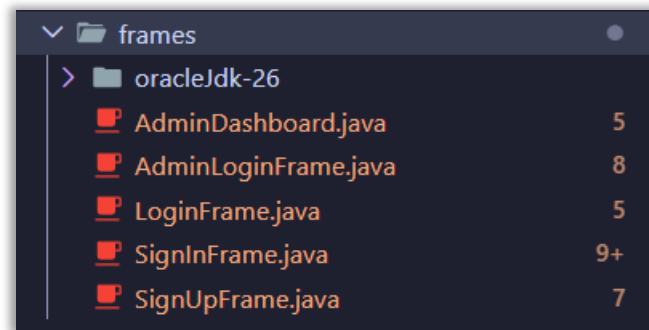


Image 38. frames Package

The ui.frames package contains the window-level frames including LoginFrame, SignUpFrame, SignInFrame, AdminLoginFrame, AdminDashboard, and the password-related frames. The main package contains RegistrationSession, a singleton class that holds all in-progress registration data in memory during the signup flow, ensuring data is only committed to the database upon final submission.



9.5 Error Handling

Error handling is implemented at both the data access layer and the user interface layer. Every DAO method wraps its database operations in a try-catch block catching `SQLException`. On failure, the error message is printed to the standard error stream using `System.err.println()` with a tag identifying the DAO and method, and the method returns false or null to the caller. This prevents database exceptions from propagating up to the UI and crashing the application.

At the UI layer, form save handlers use try-catch blocks to handle parsing errors. Date fields catch `IllegalArgumentException` thrown by `Date.valueOf()` and display a descriptive message to the user indicating the correct format. Income and numeric fields catch `NumberFormatException` when parsing `BigDecimal` or `Integer` values. Admin dialog save actions wrap all field parsing and DAO calls in a broad try-catch that displays the caught exception's message in an error dialog, ensuring the administrator always receives feedback on what went wrong.

Input validation acts as the first line of defense before any database operation is attempted. Required fields are checked for emptiness, combo boxes are checked against their default placeholder, and date fields are validated for format, logical correctness, and future-date restrictions before `Date.valueOf()` is even called. This layered approach, validate first, parse second, catch exceptions third, minimizes the likelihood of runtime errors reaching the user.



10. Conclusion

The Pag-CONNECT Digital Registration System successfully demonstrates the design and implementation of a desktop application that digitizes the Pag-IBIG Fund membership registration process. The system supports key functionalities including member registration, MID generation, employment and heirs information management, user account maintenance, and administrative oversight, all supported by a structured MySQL database.

The development of the project provided valuable experience in applying object-oriented programming concepts to a real-world system. Encapsulation, inheritance, polymorphism, and abstraction were effectively utilized to create a modular, maintainable, and scalable application. The project also reinforced the importance of layered software architecture, input validation, exception handling, and database integration in developing reliable software solutions.

Furthermore, the development process exposed the group to practical challenges such as adapting to database schema changes, managing application state across multiple interfaces, and maintaining consistency throughout the system. These experiences strengthened the group's understanding of software development practices and highlighted the value of careful planning, modular design, and clear separation of responsibilities in building effective information systems.